Transformers I

CS 4391 Introduction Computer Vision

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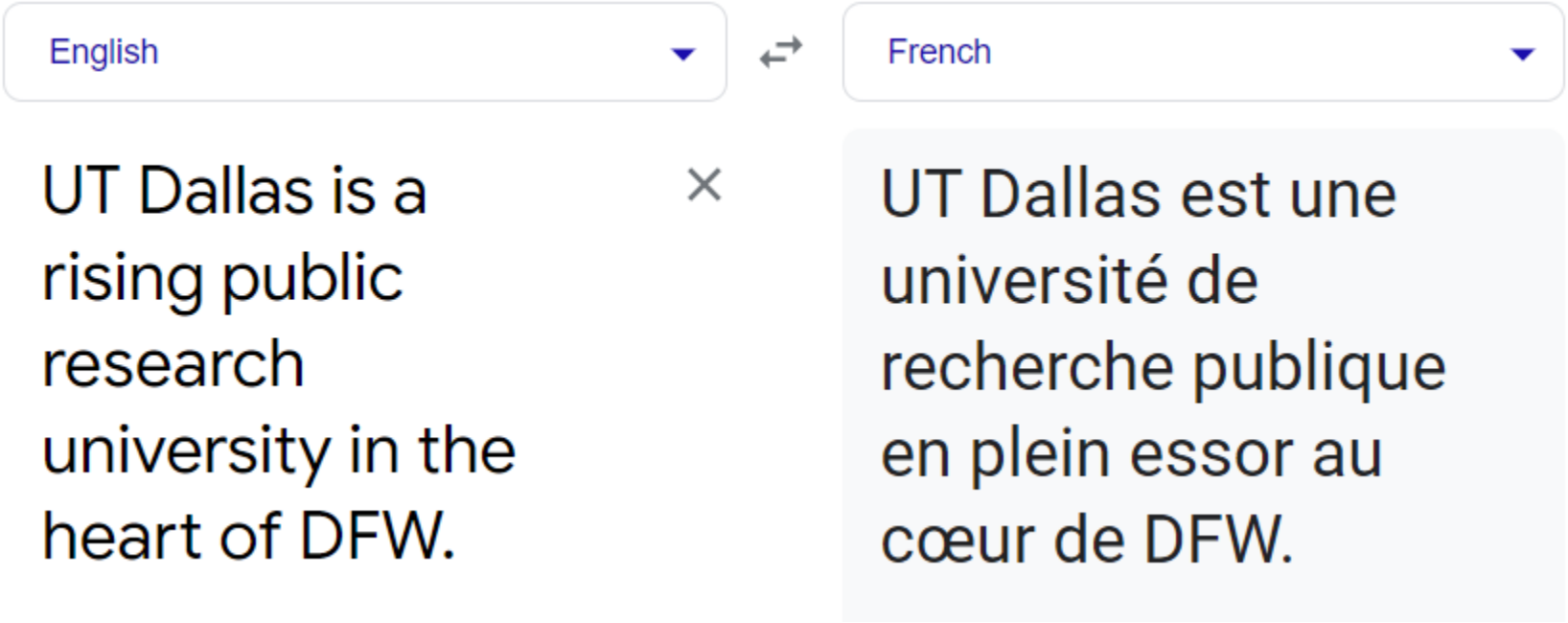
Machine Translation

- Translate a phrase from one language to anther
  - E.g., English phrase to French phrase

Google Translation

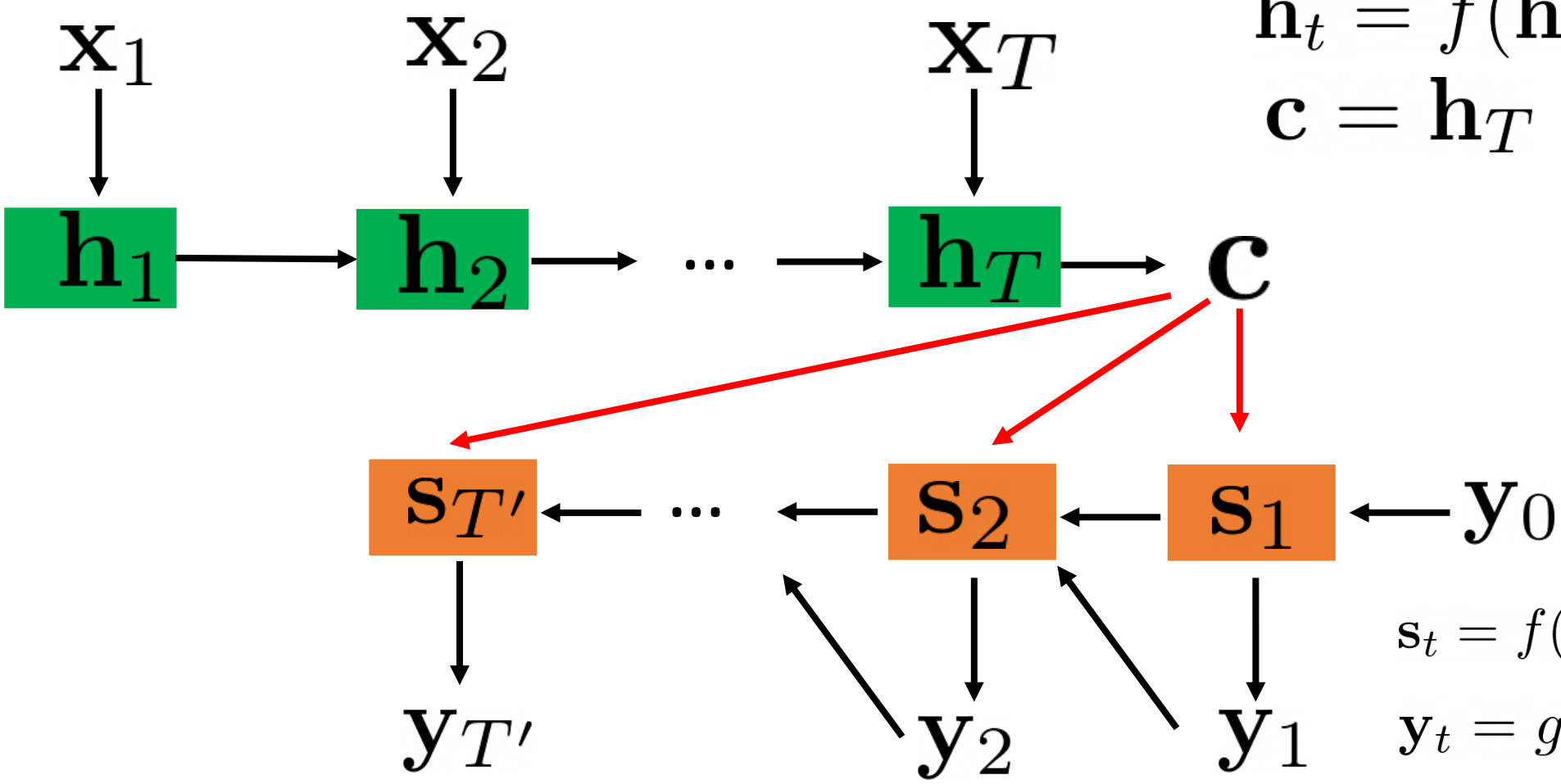


13 words

15 words

RNN Encoder-Decoder

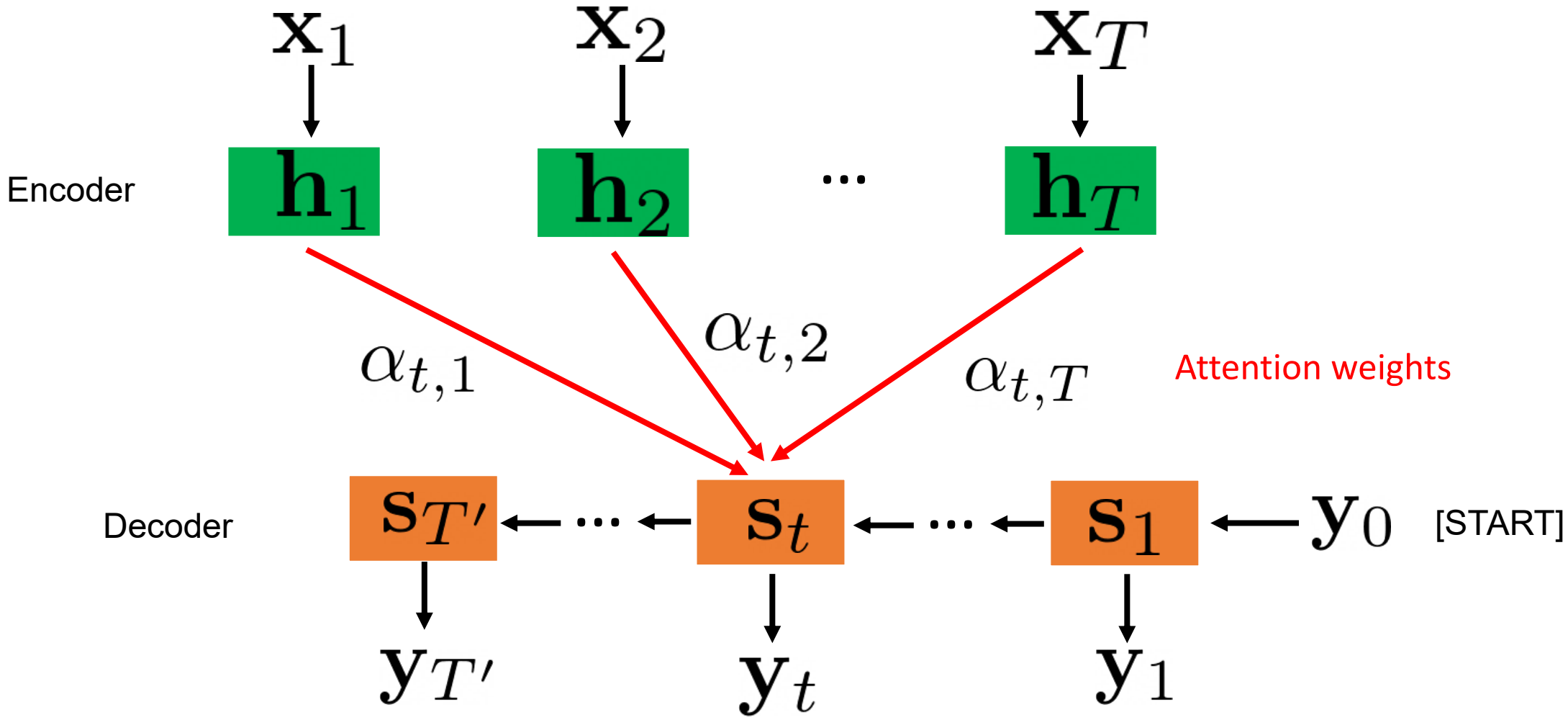
$$\mathbf{h}_t = f(\mathbf{h}_{t-1}, \mathbf{x}_t)$$
$$\mathbf{c} = \mathbf{h}_T$$



$$\mathbf{s}_t = f(\mathbf{s}_{t-1}, \mathbf{y}_{t-1}, \mathbf{c})$$
$$\mathbf{y}_t = g(\mathbf{s}_t, \mathbf{y}_{t-1}, \mathbf{c})$$

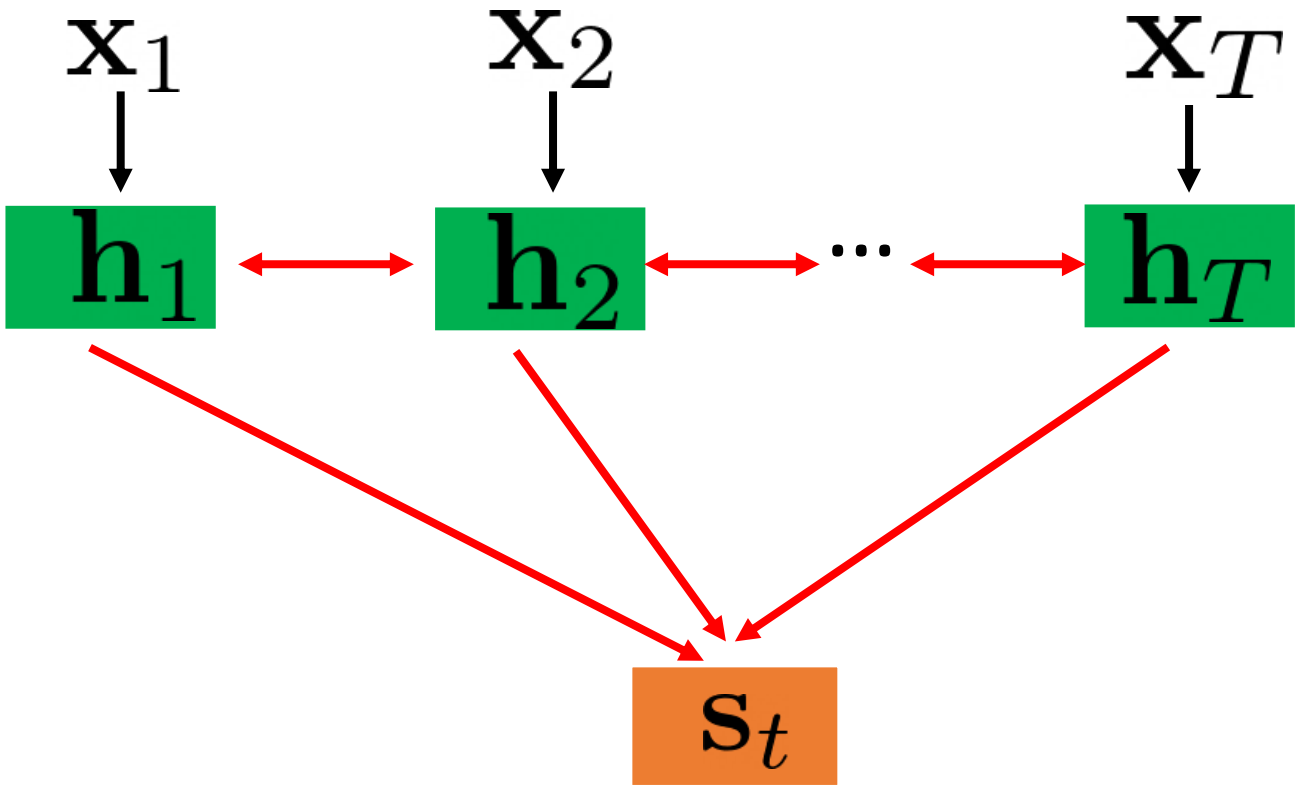
Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation. Cho et al., EMNLP'14

Transformer: Encoder-Decoder with Attention



Transformer: Attention

- Two types of attentions
- Self-attention
- Cross-attention



Transformer: Attention

- Input
  - (key, value) pairs (think about python dictionary)
  - A query

- Output
  - Compare the query to all the keys to compute weights
  - Weighted sum of the values

Attention is all you need. Vaswani et al., NeurIPS'17

Transformer: Attention

- Scaled Dot-Product Attention
  - Keys $K : m \times d_k$
  - Values $V : m \times d_v$
  - n queries $Q : n \times d_k$

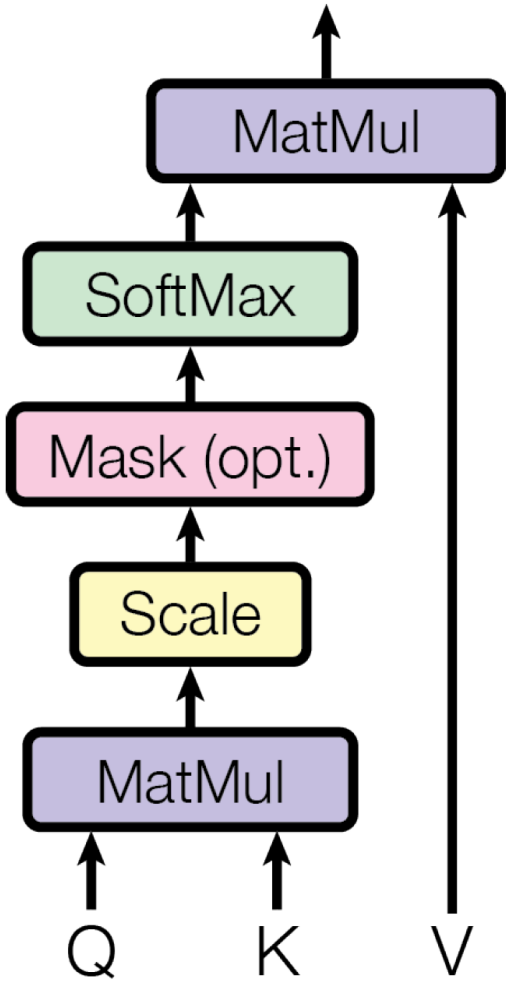
$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

$$n \times d_v$$

weights

Softmax function

$$\sigma(\mathbf{y})_i = \frac{e^{y_i}}{\sum_j^m e^{y_i}}$$



Attention is all you need. Vaswani et al., NeurIPS'17

Transformer: Attention

- Multi-Head Attention
  - Suppose the latent vector is with dimension d_{model}

$$m \times d_{\text{model}} \quad d_{\text{model}} \times d_k$$

$$\text{head}_\text{i} = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

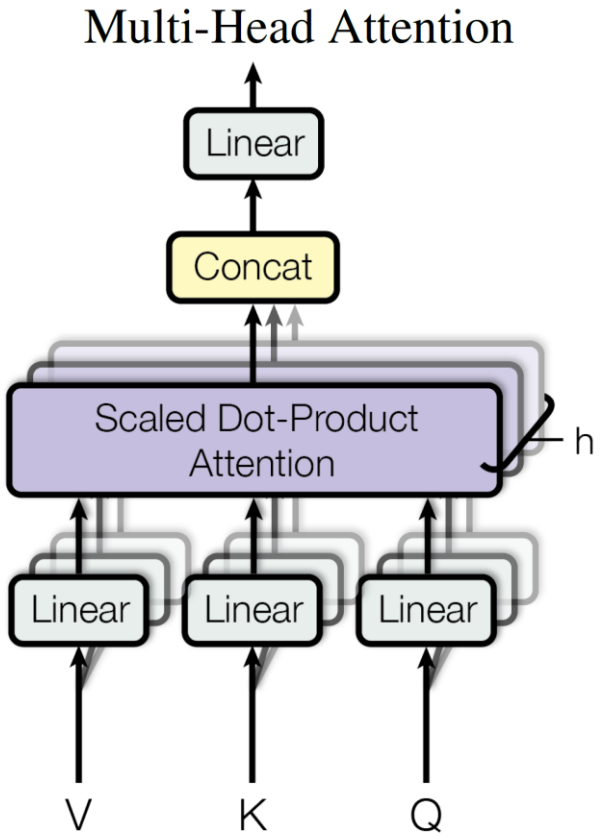
Projection

$$n \times d_v$$

$$n \times d_{\text{model}} \quad d_{\text{model}} \times d_k \qquad m \times d_{\text{model}} \quad d_{\text{model}} \times d_v$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, ..., \text{head}_\text{h})W^O$$

$$n \times d_{\text{model}} \qquad n \times hd_v \qquad hd_v \times d_{\text{model}}$$



Attention is all you need. Vaswani et al., NeurIPS'17

Transformer: Encoder

- Self-attention (repeat N times)
  - Keys, values and queries are all the same
  - n input tokens $n \times d_{\text{model}}$

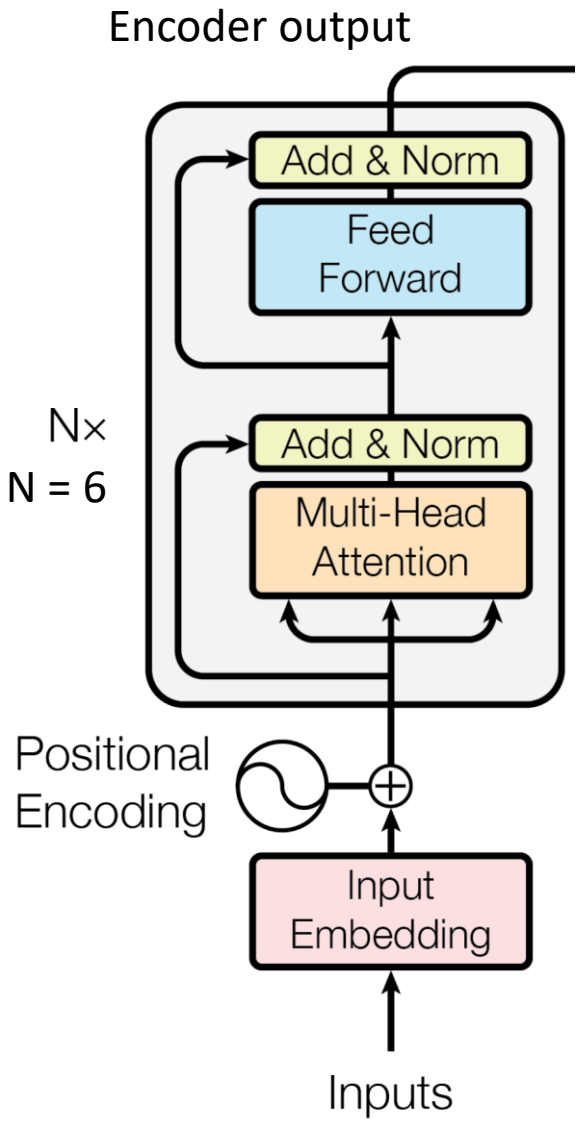
$$\text{MultiHead}(Q, K, V)$$

  - Residual connection

$$\text{LayerNorm}(x + \text{Sublayer}(x))$$

  - Layer normalization

$$\mu^l = \frac{1}{H}\sum_{i=1}^{H} a_i^l \qquad \sigma^l = \sqrt{\frac{1}{H}\sum_{i=1}^{H}\left(a_i^l - \mu^l\right)^2} \qquad \frac{a^l - \mu^l}{\sigma^l}$$



Attention is all you need. Vaswani et al., NeurIPS'17

Transformer: Encoder

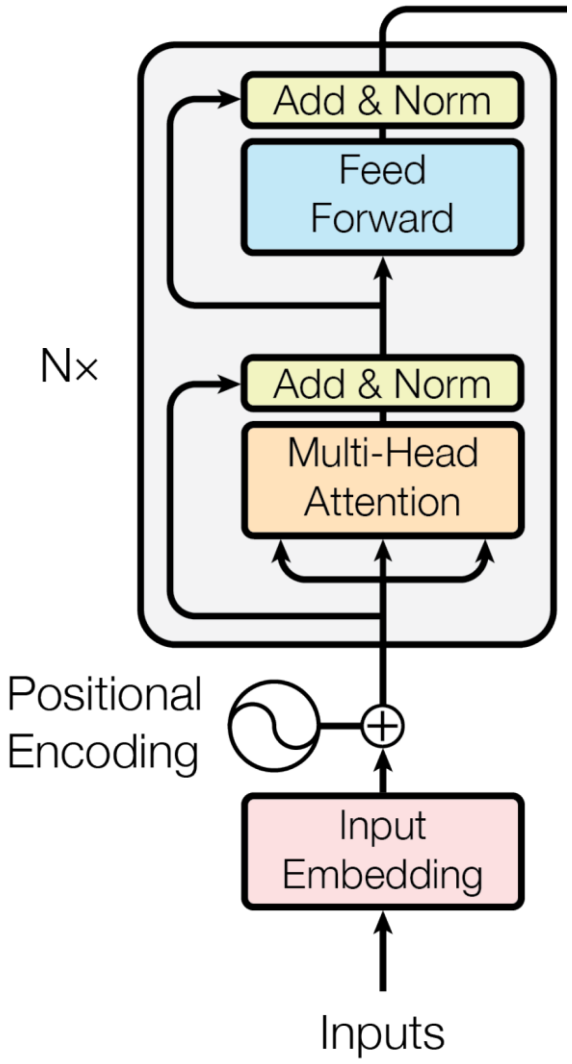
- Feed Forward Network

$$\mathrm{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

- Positional encoding
  - Make use the order of the sequence
  - With dimension d_{model} for each input

$$PE_{(pos,2i)} = sin(pos/10000^{2i/d_{\mathrm{model}}})$$

$$PE_{(pos,2i+1)} = cos(pos/10000^{2i/d_{\mathrm{model}}})$$



Attention is all you need. Vaswani et al., NeurIPS'17

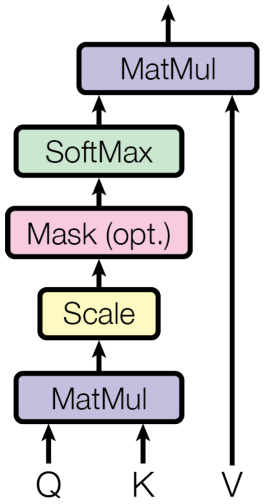
Transformer: Decoder

- Output embedding

[START]

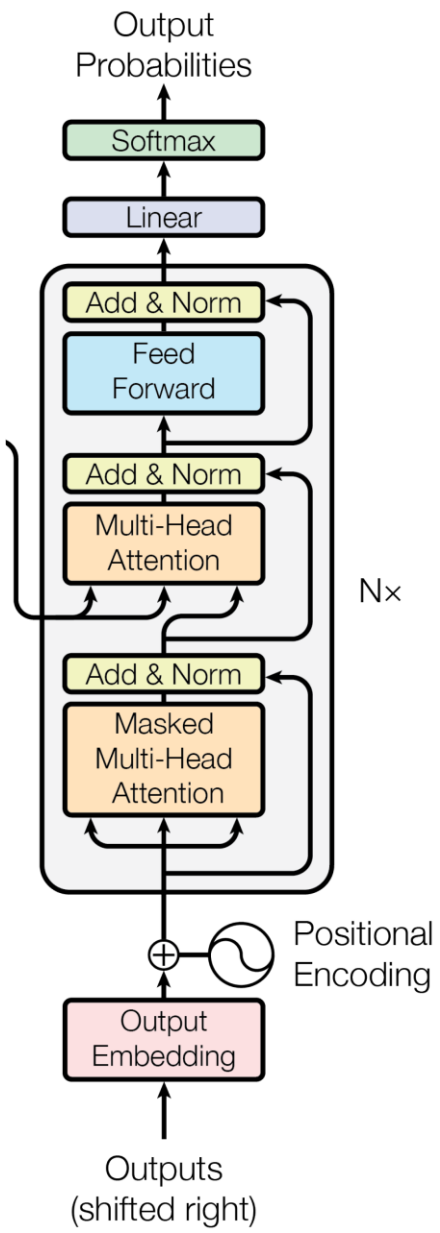
$$\mathbf{y}_0 \ \mathbf{y}_1 \cdots \mathbf{y}_{t-1} \boxed{\mathbf{y}_t \ \ \mathbf{y}_{t+1} \ \cdots \ \mathbf{y}_{T'}}$$

Shifted right by one position and insert the start token



Mask out current and future outputs during training (setting to $-\infty$)

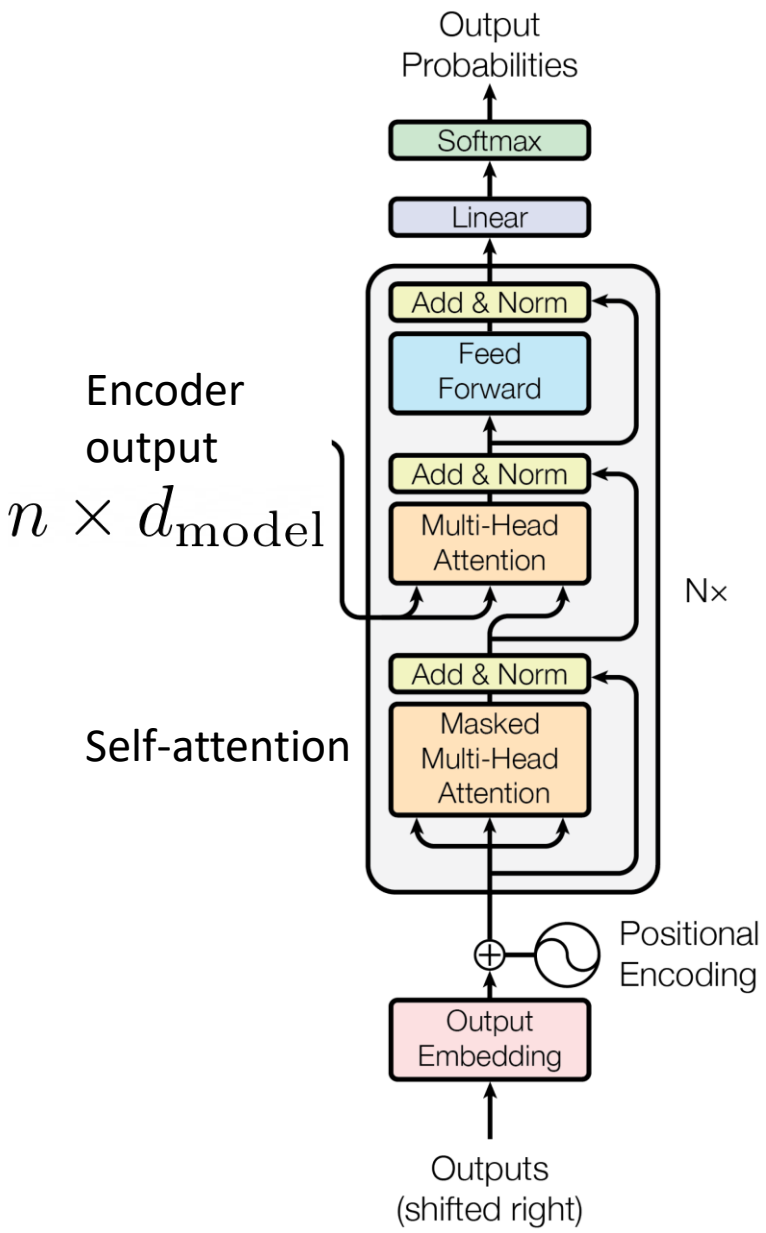
Encoder output $n \times d_{\text{model}}$



Attention is all you need. Vaswani et al., NeurIPS'17

Transformer: Decoder

- Encoder-decoder attention
  - (Key, value): encoder output
  - Queries: decoder output
  - Every position in the decoder attends to all positions in the input sequence
- Softmax
  - Predicts next-token probabilities



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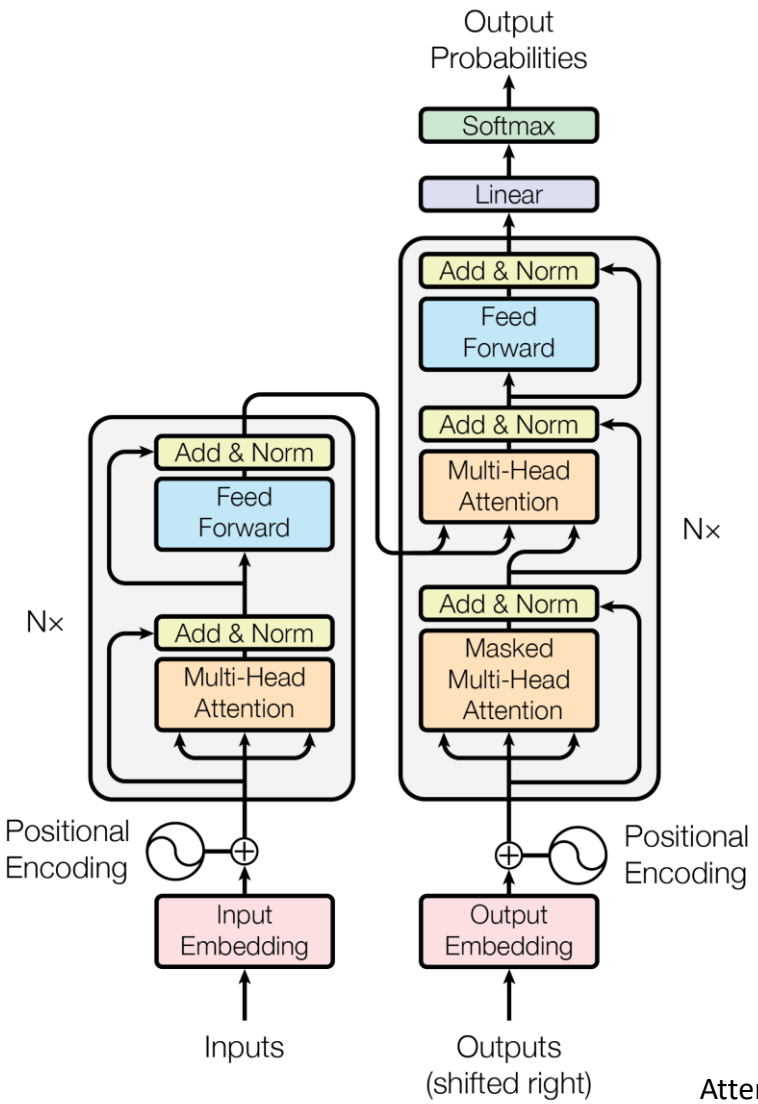
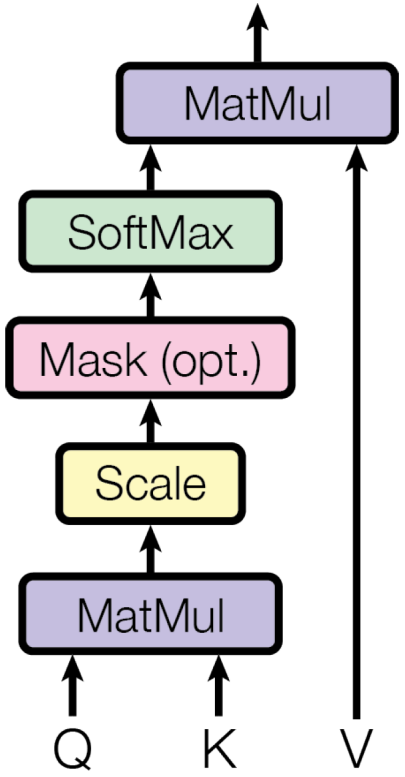
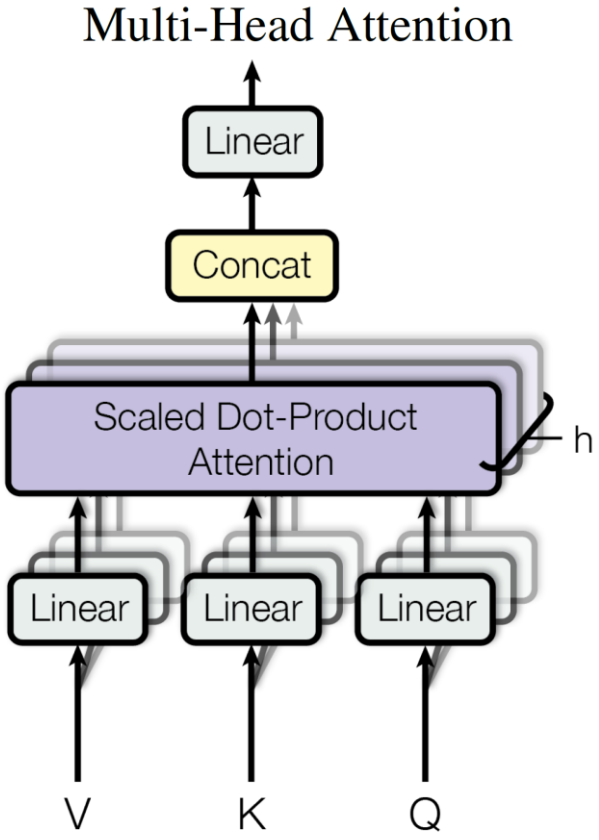
Transformer







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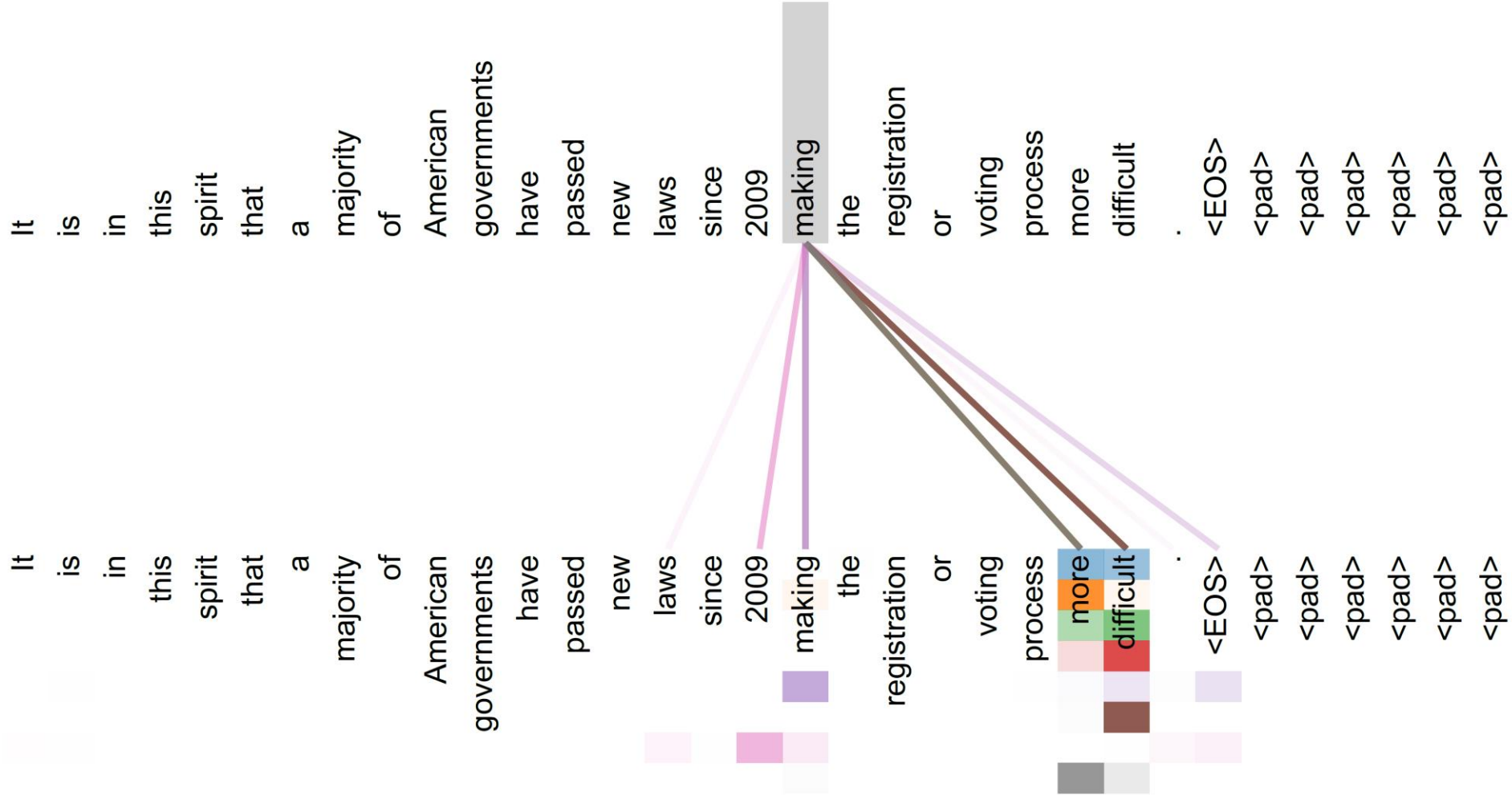
Transformer: Attention Visualization



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Summary

- Transformers
  - Can capture long-distance dependencies (global attention)
  - Computationally efficient, more parallelizable

Further Reading

- Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation https://arxiv.org/abs/1406.1078

- Neural Machine Translation by Jointly Learning to Align and Translate https://arxiv.org/abs/1409.0473

- Transformer: Attention is all you need https://arxiv.org/abs/1706.03762